Human Immunodeficiency Virus—Associated Cerebral Aneurysmal Vasculopathy: A Systematic Review

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Key words
- Cerebral aneurysm
- HIV-associated cerebral vasculopathy
- Human immunodeficiency virus
- Subarachnoid hemorrhage

Abbreviations and Acronyms
ACA: Anterior cerebral artery
ART: Antiretroviral therapy
HAART: Highly active antiretroviral therapy
HIV: Human immunodeficiency virus
SAH: Subarachnoid hemorrhage
VZV: Varicella-zoster virus

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BACKGROUND: **Human immunodeficiency virus (HIV)—associated cerebral aneurysmal vasculopathy is a rare complication of HIV affecting pediatric and adult patients and has been the subject of many case reports and case series.**

METHODS: **We performed a systematic literature search of PubMed, EMBASE, Scopus, Web of Science, Science Direct, and Google Scholar up to April 10, 2015. Our inclusion criteria encompassed all reported original case series and reports of HIV-associated cerebral aneurysms diagnosed radiologically. We analyzed the clinical characteristics and management of the reported cases.**

RESULTS: **We identified 61 patients reported in the literature (45 pediatric and 16 adult patients). The median age was 9.8 years for pediatric patients and 36.5 years for adult patients. Weakness was the most common presenting symptom in adult and pediatric patients. The most common affected artery was the middle cerebral artery. Approximately 87.2% of pediatric patients and 42.9% of adult patients were on antiretroviral therapy (ART) at presentation. The mortality rate was 60% and 35.7% among pediatric and adult patients, respectively. The optimal management is not well established. Variable response to ART was reported with possible survival benefits when ART was initiated early.**

CONCLUSIONS: **HIV-associated cerebral aneurysmal arteriopathy is associated with high mortality. The optimal management is not well established, but early initiation of ART may improve the survival rate.**

INTRODUCTION

Human immunodeficiency virus (HIV)—associated cerebral aneurysmal vasculopathy is a rare condition that has been reported in patients with HIV infection since the late 1980s.[1,2] It is typically characterized by diffuse fusiform aneurysmal dilations of the cerebral vessels without evidence of a secondary cause, such as opportunistic infections.[3,4] HIV-associated cerebral vasculopathy has been the subject of several case reports and case series. The first case was reported in 1989 in a pediatric patient who presented with weakness and aphasia.[2] Later in 2006, the first adult case of this condition was reported.[5] Authors have postulated many mechanisms to explain the etiology of these intracranial aneurysms; however, the exact mechanism remains unknown.[4,6,7] Patients usually present with weakness and radiologic evidence of infarction or subarachnoid hemorrhage (SAH).[1,8,9] Many patients with HIV-associated cerebral vasculopathy have a long period of depressed immune system and low CD4 counts.[4,5,10] The optimal management of HIV-associated cerebral aneurysmal vasculopathy is still unclear.[4] Although aneurysms were managed with conservative measures only in most patients, surgical interventions were reported in other patients.[10-12] We conducted a systematic review of all pediatric and adult cases of HIV-associated cerebral vasculopathy reported in the literature to identify the clinical characteristics and outcomes of these patients.

MATERIALS AND METHODS

Search Strategy

An online systematic review of literature was performed using PubMed, EMBASE, Scopus, Web of Science, Science Direct, and Google Scholar databases up to April 10, 2015. Articles resulting from these searches and relevant references cited in those articles were reviewed; studies included were not limited by publication date or country. The search was performed using the following search terms: "HIV and intracranial aneurysm," "HIV and intracerebral aneurysm," "HIV and fusiform aneurysm," "HIV and vasculopathy," "HIV-associated vasculopathy," and "HIV and arteriopathy."

Inclusion and Exclusion Criteria

All patients with HIV who had cerebral aneurysms consistent with HIV-associated cerebral vasculopathy were included. Patients with aneurysms not consistent with HIV-associated cerebral vasculopathy and patients with non—HIV-related secondary causes were excluded. Patients with HIV infection with cerebral or vascular changes (e.g., ischemia, hemorrhage, thrombosis, stenosis) without cerebral aneurysms were also excluded. A flow chart summarizing our search strategy is presented in **Figure 1**.

Data Collection

Data were collected using a standardized data collection form. Collected data included age, sex, source of HIV infection, number of years with infections before aneurysm diagnosis, presentation, CD4 count, viral load, vessels involved, site of involvement, type of aneurysm, radiologic findings, receipt of antiretroviral medications before aneurysm diagnosis, management, outcome, and follow-up duration.

Statistical Analysis

Data were analyzed using IBM SPSS Statistics version 20 (IBM Corporation, Armonk, New York, USA). Only reports that contained adequate clinical information were included in the analysis. To compare categorical variables, χ^2 testing was used. Independent t test was used to compare means. Level of statistical significance was determined at $P < 0.05$.

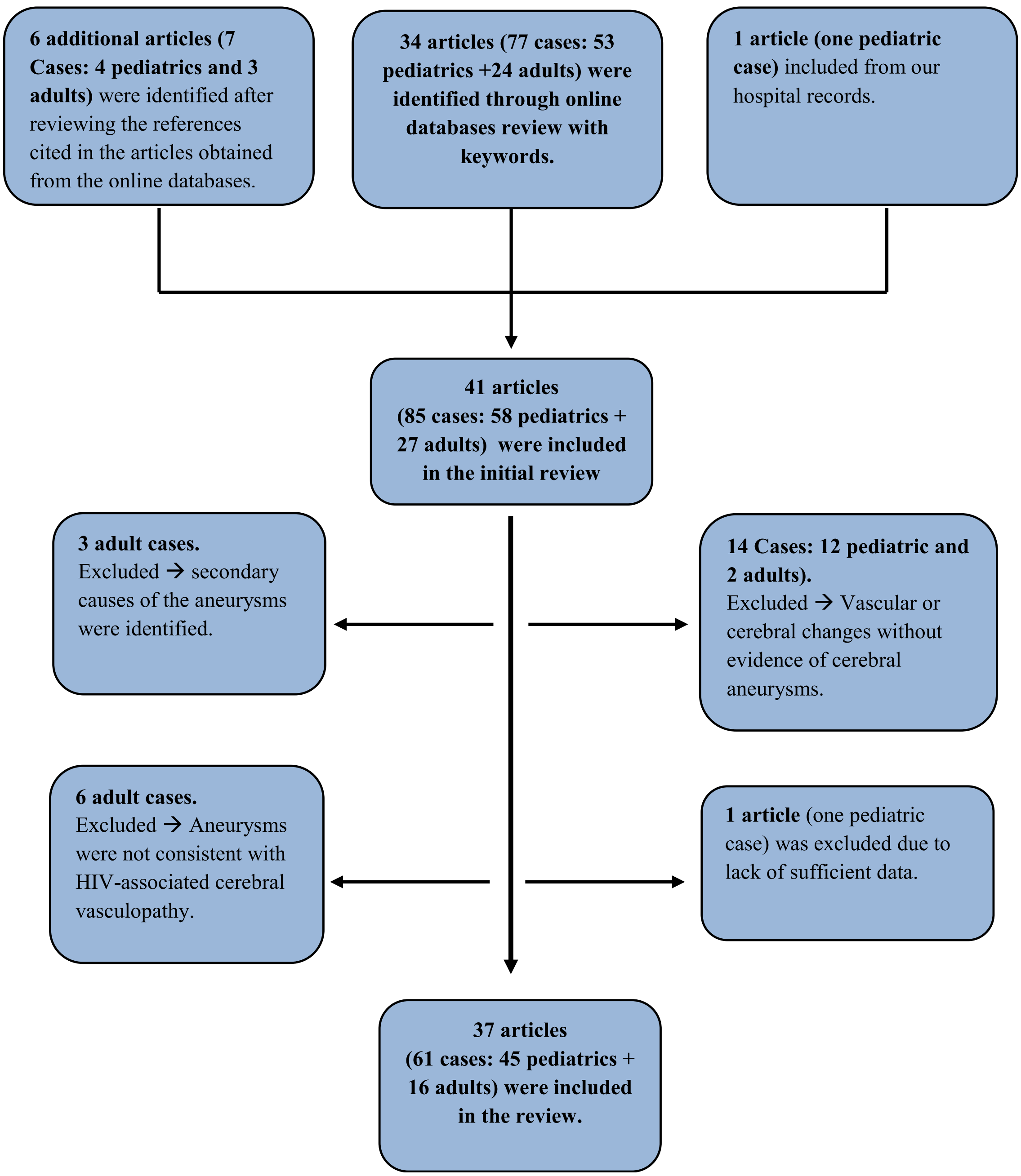

Figure 1. Flow chart of 61 cases identified and included in our review of pediatric and adult cases of human immunodeficiency virus–associated cerebral vasculopathy.

RESULTS

We identified 24 reports describing 45 pediatric cases and 13 reports describing 16 adult cases of HIV-associated cerebral vasculopathy. The basic demographic and clinical characteristics of th e cases are summarized in Tables 1 and 2. Table 3 summarizes the results of our review.

The pediatric patients ranged in age from 2 months to 18 years with a median age of 9.8 years. Male patients accounted for 65.8% of the reported cases. The source of HIV infection was identified in 31 pediatric cases. Most of the patients were infected through the vertical route ($n = 22$; 71%), and the remaining cases ($n = 9$; 29%) were infected through blood transfusion; 3 (33%) of the 9 cases acquired via blood transfusion had HIV infection during hospitalization in an early neonatal period. The mean duration of infection before the diagnosis of the aneurysm among the reported pediatric cases was 8 years (range, 0–12.6 years). The age range of the adult patients was 20–54 years with a median of 36.5 years. Male patients accounted for 56.3% of the reported cases. The mean duration of infection before the diagnosis of the aneurysm was 4.6 years (range, 0–19 years).

Weakness was the most common presenting symptom among pediatric and adult patients (65.6% and 50%, respectively). Convulsions (21.9%) followed by headache (18.8%) were the next most common presenting symptoms among pediatric patients, whereas headache (31.3%), confusion (25%), and speech disorders (25%) were the next most common presenting symptoms in adult patients (Table 3). Most of the aneurysms were multiple (85% of pediatric cases and 93.8% of adult cases). Fusiform aneurysms were reported in 86.7% of pediatric cases and 75% of adult cases. Saccular aneurysms were reported in 8.9% of pediatric cases and 12.5% of adult cases. One pediatric case (4.4%) and one adult case (12.5%) had both saccular and fusiform aneurysms (Table 3).

The commonly affected arteries in adults were the middle cerebral artery (MCA) followed by the anterior cerebral artery (ACA) (68.8% and 62.5% of cases, respectively). The internal carotid artery and the posterior cerebral artery were equally affected (37.5% of cases). Among pediatric cases, the MCA was the most commonly affected artery (70.6% of cases) followed by the internal carotid artery (67.7%) and the ACA (61.8%) (Table 3).

Among patients with cerebral aneurysms secondary to HIV-associated cerebral vasculopathy, 36.4% of pediatric patients and 37.5% of adult patients had ischemic cerebral infarction at the time of diagnosis of the aneurysms, whereas 45.5% of pediatric patients and 25% of adult patients had aneurysms without ischemia or hemorrhage (Table 3). Brain hemorrhage was reported in 9.1% of pediatric patients and 43.8% of adult patients.

CD4 cell count results were available for only 24 of 45 pediatric cases. The CD4 cell count was 0–646 cells/mm^3 with a median of 43 cells/mm^3. Only 7 cases (29.1%) had CD4 cell counts >200 cells/mm^3. The CD4 cell count at the time of diagnosis of the aneurysms in the 16 adult patients was 0–496 cells/mm^3 with a median of 76 cells/mm^3. Only 3 cases (18.75%) had CD4 counts >200 cells/mm^3, the defining cutoff count of stage 4 (acquired immunodeficiency syndrome) of HIV infection.

Of the pediatric patients, 34 (87.2%) were receiving antiretroviral therapy (ART) at the time of the diagnosis of the aneurysms, 5 (12.8%) patients were not receiving treatment, and the remaining 6 cases had missing data. The group receiving treatment included patients who were on ART with 1 or 2 antiretroviral medications and patients on highly active antiretroviral therapy (HAART) regimens using combination of at least 3 medications. Only 6 (42.9%) adult patients were receiving ART at presentation, 8 (57.1%) were not receiving treatment, and the remaining 2 cases had missing information.

At presentation, 5 (83.3%) of 6 adult patients receiving ART and 9 (24.3%) of 34 pediatric patients receiving ART were on HAART therapy. The lower proportion of pediatric cases treated with HAART is explained by the fact that most pediatric cases were reported before the era of HAART. The outcome of patients who were on HAART therapy was better than the outcome of patients who received 1- or 2-drug regimens. Of 9 (77.8%) pediatric patients who were on HAART, 7 were alive compared with only 11 (44%) of 25 patients who received 1- or 2-drug regimens ($P = 0.086$). Additionally, the mean CD4 counts of adult patients on therapy was higher compared with patients who were not on therapy (193.2 cells/mm^3 vs. 88.5 cells/mm^3; $P = 0.175$).

Most pediatric patients ($n = 41$; 91.1%) were managed conservatively, including admission to intensive care units, prophylactic antibiotics and antiviral agents, initiation of ART, and conservative management of infarction and hemorrhage without any surgical intervention or invasive procedure. Only 4 cases underwent an invasive or a surgical procedure. In 1 case, surgical evacuation of an intracerebral hemorrhage was performed.[18] In another case, an external ventricular drain was inserted to treat hydrocephalus.[29] A twist drill with ventricular tapping was used in a third case.[4] In a fourth case, a patient presented with SAH and hydrocephalus that was treated with insertion of an external ventricular drain and subsequently underwent a craniotomy with surgical clipping of the aneurysm. In the adult group, 12 (80%) of 15 patients with complete data received conservative management only. The other 3 (20%) patients underwent surgical or endovascular interventions; the first patient underwent an unsuccessful attempt at repairing a fusiform aneurysm[6]; the second patient underwent craniotomy with surgical clipping of the aneurysm[12]; the third case underwent an endovascular treatment with flow-diverting stents.[11]

During the follow-up period, 27 (60%) of 45 pediatric patients died and 18 (40.0%) survived. The mean duration of follow-up was 1.3 years (range, 0–3.6 years). The cause of death was identified in 22 of the 27 patients who died. Only 8 (36.4%) patients died as a result of neurologic complications, and the remaining patients ($n = 14$; 63.6%) died from other causes, including HIV-related opportunistic infections and other complications of immunodeficiency. The mean follow-up duration before death was 0.97 years. The mean CD4 count of the pediatric patients who survived was 193 cells/mm^3 compared with 25 cells/mm^3 of the pediatric patients who died ($P = 0.033$).

Table 1. Pediatric Cases of Human Immunodeficiency Virus—Associated Cerebral Vasculopathy

Serial Number	Author	Age (years)/Sex	Source of HIV	Time From Infection to CNS Event (years)	ART on Presentation, Years on Treatment	Number of Aneurysms, Type of Aneurysms	Imaging Findings Other Than Aneurysm	Management	Outcome
1	Kure et al.[2]	5.7/M	Perinatal	5.7	No, 0	Multiple, fusiform	Infarction	Conservative	Dead
2	Frank et al.[13]	4.5/F	Perinatal	4.5	No, 0	Multiple, fusiform	Infarction	Conservative	Dead
3	Park et al.[14]	6/M	Perinatal	6	—	Multiple, fusiform	Infarction	Conservative	Dead
4	Lang et al.[15]	8/M	Perinatal	8	Yes, 5	Multiple, fusiform	None	Conservative	Dead
5	Husson et al.[16]	11/M	Blood transfusion	11	Yes, 2	Multiple, fusiform and saccular	None	Conservative	Alive
6		12/M	Blood transfusion	6	Yes, 3	Multiple, fusiform	None	Conservative	Alive
7	Philippet et al.[17]	9/F	Perinatal	9	Yes, 4	Multiple, fusiform	Infarction	Conservative	Alive
8		14/M	Blood transfusion	3	Yes, 1.5	Multiple, fusiform	Infarction	Conservative	Dead
9	Shah et al.[18]	18/M	Blood transfusion	6	—	Multiple, fusiform	Intracerebral hemorrhage	Surgical evacuation of hematoma	Dead
10		12/F	Perinatal	12	—	Multiple, fusiform	Infarction	Conservative	Dead
11	Dubrovsky et al.[10]	12.6/M	Unknown	—	Yes, 1.5	Multiple, fusiform	Infarction	Conservative	Dead
12		11/F	Perinatal	11	Yes, 1	Multiple, fusiform	Infarction	Conservative	Dead
13		6/F	Perinatal	6	Yes, 4	Multiple, fusiform	Lymphoma	Conservative	Dead
14		12/M	Blood transfusion	12	Yes, 2	Multiple, fusiform	Infarction	Conservative	Dead
15		10/M	Perinatal	10	Yes, 2	Multiple, fusiform	Not done	Conservative	Dead
16	Fulmer et al.[19]	11/F	Perinatal	11	—	Multiple, fusiform	Infarction	Conservative	Dead
17	Mazzoni et al.[8]	8.3/F	Perinatal	8.3	Yes, 8	Multiple, fusiform and saccular	Hemorrhage and hydrocephalus	Conservative	Alive
18	Carvalho et al.[20]	6.5/M	Perinatal	6.5	Yes, 4	Multiple, fusiform	None	Conservative	Alive
19	Nunes et al.[21]	0.17/F	Perinatal	0.17	Yes, 0.17	Single, saccular	Hemorrhage and hydrocephalus	Conservative	Dead
20	Elfenbein and Emmanuel[22]	8/M	Unknown	—	Yes, 2.5	Multiple, fusiform	None	Conservative	Alive
21	Visrutaratna and Oranratanachai[23]	6/F	Perinatal	6	No, 0	Multiple, fusiform	None	Conservative	—
22	Bonkowsky et al.[24]	12/M	Perinatal	12	Yes, —	Multiple, fusiform	Infarction	Conservative	Alive
23	Patsalides et al.[25]	7.5/M	Perinatal	7.5	Yes, 5	5 fusiform and 2 saccular aneurysms; 4 patients had multiple aneurysms and 3 had single aneurysms	Infarction	Conservative	Alive
24		9.3/F	Perinatal	9.3	Yes, 7.3		Hydrocephalus	Conservative	Dead
25		9.4/M	Blood transfusion	1.9	Yes, 1.9		Infarction	Conservative	Dead
26		10.5/M	Blood transfusion	2.6	Yes, 2.6		None	Conservative	Dead
27		15.1/M	Blood transfusion	0.83	Yes, 0.8		None	Conservative	Dead
28		12.6/F	Perinatal	12.6	Yes, 3.8		Infarction	Conservative	Dead
29		8.4/M	Perinatal	8.4	Yes, 5.8		Infarction	Conservative	Alive
30	Petropoulou et al.[7]	12/M	Blood transfusion	12	Yes, —	Multiple, saccular	None	Conservative	Alive

Continues

Table 1. Continued

Serial Number	Author	Age (years)/Sex	Source of HIV	Time From Infection to CNS Event (years)	ART on Presentation, Years on Treatment	Number of Aneurysms, Type of Aneurysms	Imaging Findings Other Than Aneurysm	Management	Outcome
31	Martínez-Longoria et al.[26]	12/F	Perinatal	12	Yes, —	Multiple, fusiform	None	Conservative	Alive
32	Mahadevan et al.[4]	16/M	Unknown	—	—	Multiple, fusiform	Hydrocephalus	Twist drill with ventricular tapping	Dead
33	Thakker and Bahatia[27]	12/M	Perinatal	12	Yes, 0.3	Single, fusiform	None	Conservative	Alive
34	Demopolous et al.[28]	12/M	Unknown	—	Yes, 5	Single, fusiform	None	Conservative	Alive
35		6/F	Unknown	—	Yes, 0.5	Multiple, fusiform	None	Conservative	Alive
36		7/M	Unknown	—	No, 0	Multiple, fusiform	None	Conservative	Dead
37	Savitr Sastri et al.[29]	13/M	Unknown	—	—	Multiple, fusiform	Hydrocephalus	External ventricular drain was inserted	Dead
38	Schieffelin et al.[30]	—		—	Yes, —	Multiple, fusiform	None	Conservative	Alive
39		—		—	Yes, —	Multiple, fusiform	Infarction	Conservative	Dead
40		—		—	Yes, —	Single, fusiform	None	Conservative	Dead
41		—		—	Yes, —	Multiple, fusiform	None	Conservative	Alive
42		—		—	Yes, —	Multiple, fusiform	None	Conservative	Alive
43		—		—	Yes, —	Multiple, fusiform	None	Conservative	Alive
44		—		—	Yes, —	Multiple, fusiform	None	Conservative	Dead
45	Bakhaidar et al.[31]	7/M	Perinatal	7.00	No, 0	Multiple, fusiform	Hemorrhage and hydrocephalus	Surgical clipping of aneurysm	Dead

HIV, human immunodeficiency virus; CNS, central nervous system; ART, antiretroviral; M, male; female.

Among 14 of 16 adult cases with complete data, 5 (35.7%) patients died and 9 (64.3%) patients survived during the follow-up period. The mean follow-up duration was 1.4 years. The cause of death was identified in 4 of the 5 patients who died; 2 (50%) patients died as a result of neurologic complications, and the other 2 (50%) patients died from other causes. The mean follow-up duration preceding death of patients was 0.87 years. The mean CD4 count of the adult patients who survived was 126.9 cells/mm^3 compared with 137.2 cells/mm^3 among the adult patients who died.

DISCUSSION

HIV infection is well recognized to be associated with cerebrovascular complications in adult and pediatric patients.[10,39] Ischemic stroke is the most commonly reported cerebrovascular complication of HIV infection.[29] Other cerebrovascular complications related to HIV are moyamoya disease and HIV-associated cerebral vasculopathy in the form of intracerebral aneurysms.[39-41] The true incidence of cerebrovascular complications in HIV infection is difficult to estimate because of the high rate of association with opportunistic bacterial, viral, and spirochetal infections of the nervous system that also involve cerebral vessels.[41] HIV-associated cerebral aneurysmal vasculopathy, also termed HIV-associated cerebral arteriopathy and previously termed cerebral aneurysmal childhood arteriopathy, is a rare manifestation of HIV infection that occurs more frequently in children infected with HIV.[5,10,27,29] It is characterized by the formation of multiple intracranial fusiform aneurysms usually in arteries of the circle of Willis.[10,16] In the present review, we summarized the clinical characteristic of all cases of this rare condition cases. Vertical transmission (i.e., mother-to-child transmission during pregnancy or delivery or postpartum via breastfeeding) is responsible for >90% of HIV infection in children.[42] Most reported pediatric cases with HIV-associated cerebral vasculopathy were infected through the vertical route. Among adults, transmission of HIV infection through sexual contact is considered the most common route of infection.[42,43] We could not identify data on the source of infection among adult patients.

Table 2. Adult Cases of Human Immunodeficiency Virus—Associated Cerebral Vasculopathy

Serial Number	Author	Age (years)/Sex	Time From Infection to CNS Event (years)	ART on Presentation, Years on Treatment	Number of Aneurysms, Type of Aneurysms	Imaging Findings Other Than Aneurysm	CD4 Count	Management	Outcome
1	Ake et al.[5]	29/F	9	Yes, —	Multiple, fusiform	SAH and IVH	2	Conservative	Dead
2	Kossorotoff et al.[32]	20/M	19	Yes, —	Multiple, fusiform	Infarction	496	Conservative	Dead
3		32/M	0.3	Yes, 0.3	Multiple, fusiform	Infarction	338	Conservative	Alive
4	Miyamoto et al.[12]	54/M	0.2	No	Multiple, saccular	SAH and IVH	112	Surgical clipping of aneurysm	Alive
5	Tipping et al.[9]	27/F	—	—	Multiple, fusiform	Infarction	14	Conservative	Dead
6	Hamilton et al.[33]	34/M	10	Yes, 0.5	Multiple, fusiform and saccular	SAH	64	Conservative	Alive
7	O'Charoen et al.[34]	36/M	0.3	No	Multiple, fusiform	Infarction	43	—	—
8	Statler et al.[35]	36/F	9	—	Multiple, fusiform	SAH	2	Conservative	Dead
9	Modi et al.[6]	37/M	—	No	Multiple, fusiform	None	164	Conservative	Lost to follow-up
10		43/M	—	No	Multiple, fusiform	SAH	172	Surgical intervention	Dead
11		43/M	—	No	Multiple, fusiform	None	117	Conservative	Lost to follow-up
12	Goldstein et al.[1]	38/F	0	No	Multiple, fusiform and saccular	Infarction and intracerebral hemorrhage	<1	Conservative	Alive
13	Sanada et al.[36]	43/F	—	No	Multiple, fusiform	None	12	Conservative	—
14	Benjilali et al.[37]	39/F	0	No	Single, saccular	None	87	Conservative	Alive
15	Stanley et al.[38]	22/F	1	Yes, 1	Multiple, fusiform	Infarction	210	Conservative	Alive
16	Delgado Almandoz et al.[11]	37/M	1.5	Yes, —	Multiple, fusiform	Infarction	49	Endovascular	Alive

CNS, central nervous system; ART, antiretroviral; female; M, male; SAH, subarachnoid hemorrhage; IVH, intraventricular hemorrhage.

Table 3. Results of the Systematic Review

	Pediatric Patients (n = 45)	Adult Patients (n = 16)
Sex	$n = 38$	$n = 16$
Male	25 (65.8%)	9 (56.3%)
Female	13 (34.2%)	7 (43.8%)
Presentation	$n = 32$	$n = 16$
Asymptomatic	3 (9.4%)	0 (0%)
Weakness	21 (65.6%)	8 (50%)
Convulsions	7 (21.9%)	1 (6.3%)
Confusion	4 (12.5%)	4 (25%)
Speech symptoms	4 (12.5%)	4 (25%)
Headache	6 (18.8%)	5 (31.3%)
Loss of consciousness	3 (9.4%)	1 (6.3%)
Blurred vision	2 (6.3%)	0 (0%)
Vomiting	2 (6.3%)	2 (12.5%)
Others	2 (6.3%)	4 (25%)
Vessels involved	$n = 34$	$n = 16$
ICA	23 (67.7%)	6 (37.5%)
MCA	24 (70.6%)	11 (68.8%)
ACA	21 (61.8%)	10 (62.5%)
PCA	2 (5.9%)	6 (37.5%)
Basilar	4 (11.8%)	5 (31.3%)
AComA	0 (0%)	2 (12.5%)
Others	2 (5.9%)	3 (18.8%)
Fusiform/saccular	$n = 45$	$n = 16$
Fusiform	39 (86.7%)	12 (75%)
Saccular	4 (8.9%)	2 (12.5%)
Both	2 (4.4%)	2 (12.5%)
Radiologic finding	$n = 44$	$n = 16$
Aneurysms alone	20 (45.5%)	4 (25%)
Hemorrhage	4 (9.1%)	7 (43.8%)
Ischemia	16 (36.4%)	6 (37.5%)
Hydrocephalus	6 (13.6%)	0 (0%)
Others	1 (2.3%)	0 (0%)

ICA, internal carotid artery; MCA, middle cerebral artery; ACA, anterior cerebral artery; PCA, posterior cerebral artery; AComA, anterior communicating artery.

HIV-associated cerebral vasculopathy seems to have a tendency to occur more in children.[6,9] Since the first pediatric case reported in 1989, all reported cases had been exclusively in the pediatric population until 2006, when the first adult case was reported.

HIV-related large vessel extracranial aneurysms are a well-described complication of HIV infection.[44,45] The most commonly affected arteries are superficial femoral, carotid, and popliteal arteries.[45,46] Most extracranial and intracranial HIV-related aneurysms are multiple and have distinct histopathologic features that distinguish them from atherosclerotic and mycotic aneurysms.[45,47] The main histopathologic features include vasculitis of the vasa vasorum and periadventitial vessels, proliferation of slit-like vascular channels, and chronic inflammation with fibrosis of the adventitia.[47] There is also associated medial fibrosis with loss and fragmentation of muscle and elastic tissue without intimal hyperplasia.[47] Similar findings were reported in cerebral aneurysms, with medial fibrosis, loss of the muscularis, destruction of the internal elastic lamina, and intimal hyperplasia being the key histopathologic feature of HIV-associated cerebral vasculopathy.[2,4,48] However, HIV-related cerebral vasculopathy is usually reported in isolation from involvement of systemic arteries.[4,16] This situation was hypothesized to be due to immune activation and release of cytokines by HIV-infected monocytes that are known for their brain tropism.[4,9,48]

The exact mechanism by which HIV causes arterial damage is unknown.[4,6,44] Several possible mechanisms warrant consideration.[4,7,18] The formation of aneurysms during advanced disease and profound immunosuppression prompted the consideration of a secondary infective mechanism.[4,45] Lodging of bacteria in an arterial wall during a period of bacteremia secondary to immunosuppression leading to secondary formation of mycotic aneurysms is one of the proposed mechanisms.[44,45] In addition, many reports have related the development of cerebral aneurysms to viral etiologies, such as varicella-zoster virus (VZV), cytomegalovirus, and herpes simplex virus.[16,21,25,38] Nevertheless, most reports could not stain or culture any organism from the affected specimens through either conventional culture or polymerase chain reaction methods.[4,44] Failure to demonstrate an infective organism in the affected vessels made this hypothesis less likely. Nonetheless, VZV vasculopathy is a well-reported condition in immunocompromised patients that could manifest with aneurysms similar to HIV-associated cerebral vasculopathy.[38,49] Cerebrospinal fluid analysis for anti-VZV immunoglobulin G and polymerase chain reaction for VZV DNA can be useful for the diagnosis.[49]

Direct involvement of HIV itself or via an immune complex mechanism in the pathogenesis of aneurysm formation comes from several observations. Kure et al.[2] successfully used immunohistochemistry in their reported case to stain the wall of affected arteries with antibodies against HIV transmembrane glycoprotein-41. HIV-1 DNA was amplified through polymerase chain reaction from the aneurysms in other reported cases.[4,10] Moreover, some authors observed that HIV-associated cerebral aneurysms are diagnosed during periods of high viral load and high levels of P24 antigen (a core viral protein that reflects active HIV replication); these observations further strengthen the hypothesis of a direct role of the virus in the pathogenesis.[4,7,8,16] Additionally, most of the reported patients had low CD4 counts and advanced HIV infection, with only 18.7% of adult cases and 29.1% of pediatric cases having a CD4 count >200 cells/mm^3 at the time of diagnosis of the aneurysms. However, other authors reported that their patients presented after the initiation of HAART and while CD4 counts were increasing.[24,28] They hypothesized that immune reconstitution inflammatory syndrome might have played a role in the pathogenesis of the aneurysms.[4,24,28]

We found that 35.6% of our reported pediatric patients and 37.5% of adult patients had ischemic cerebral infarction at the time of the diagnosis of the aneurysms. HIV infection appears to increase the risk of cerebral ischemia and hemorrhage.[39,50] Moreover, hemorrhagic ischemia was reported in 9.1% of pediatric patients and 43.8% of adult patients. This ischemia may explain the weakness with which most reported patients presented. In addition, HIV-associated cerebral vasculopathy, opportunistic infections, intracranial malignancies, and coagulation abnormalities can substantially contribute to the increased risk of stroke in HIV-infected patients.[39,50,51] About one tenth of the reported pediatric patients were diagnosed with aneurysms during routine periodic examination without symptoms.[7,16] However, cerebral aneurysms in the setting of HIV-associated cerebral vasculopathy can be the manifestation in patients with undiagnosed HIV infection.[1,6,29]

Extracranial systemic HIV-related aneurysms are usually treated surgically or through endovascular interventions.[45,52] In contrast, the multiplicity, diffuse distribution, and fusiform architecture of aneurysms of HIV-associated cerebral vasculopathy have limited neurosurgical or endovascular interventions.[4,5] Most of the reported patients received conservative management without surgical intervention. Many patients who received 1- or 2-drug ART regimens had adverse outcomes.[4,26] Variable response to HAART therapy was observed. Some authors reported progression of vasculopathy despite the initiation of HAART,[5] whereas others reported halting the disease progression[7,8,24] or complete resolution of the vasculopathy after the initiation of HAART.[26,37,53]

There are a few reports of surgical or endovascular interventions in patients with HIV-associated cerebral vasculopathy. Modi et al.[6] reported a patient who underwent an unsuccessful attempt at repairing a right MCA aneurysm. The artery was found to be fusiform during surgery and could not be repaired. Miyamoto et al.[12] reported an HIV-infected adult patient who presented with SAH and was found to have bilateral saccular ACA aneurysms. He underwent clipping of the aneurysms with no complications observed over 4.5 years of postoperative follow-up. Delgado Almandoz et al.[11] successfully repaired 3 fusiform aneurysms secondary to HIV-associated cerebral vasculopathy in an adult patient using the Pipeline Embolization Device (Covidien, Irvine, California, USA). The patient improved clinically, and no complications were encountered over 10 months of follow-up.[11] At our institution, microsurgical clipping and reconstructive repair was successful in a 7-year-old HIV-infected boy with a ruptured fusiform aneurysm involving the left internal carotid artery, MCA, and ACA.[31]

This systematic review has some limitations. Our search yielded only small case series or single case reports; this might have been due to publication bias, where only cases that underwent treatment or had unusual findings were reported. Our results may not reflect the true natural history of HIV cerebral aneurysmal vasculopathy because cerebral vascular imaging is not part of the routine work-up for HIV-positive patients. Thus, some patients with this condition may remain asymptomatic, and the condition may never be diagnosed. Thus, the natural history of this condition may be better than what is presented in this review. Given the retrospective design of many of the articles included, some of the data that we attempted to extract from the published articles were unavailable. Finally, our search was restricted to articles published in the English language.

CONCLUSIONS

HIV infection is a complex systemic disease that may affect the integrity of the cerebral vasculature. HIV-associated cerebral vasculopathy is a rare cerebrovascular complication of HIV infection with characteristic histopathologic features. Patients usually present with weakness and other neurologic symptoms, including headache, convulsions, and speech disorders. Earlier surveillance of patients with HIV and neurologic symptoms might reveal early disease. The severe immunosuppression and possibility of the presence of secondary opportunistic infection that can also affect cerebral vasculature may hinder the diagnosis of HIV-associated cerebral vasculopathy. Because of the rarity of reports of this condition in the literature, optimal management is unclear. Response to HAART therapy varied in the literature, with some reports demonstrating survival benefits when HAART is initiated early. Surgical or endovascular management of ruptured or progressing HIV-associated cerebral aneurysms is recommended regardless of HAART.

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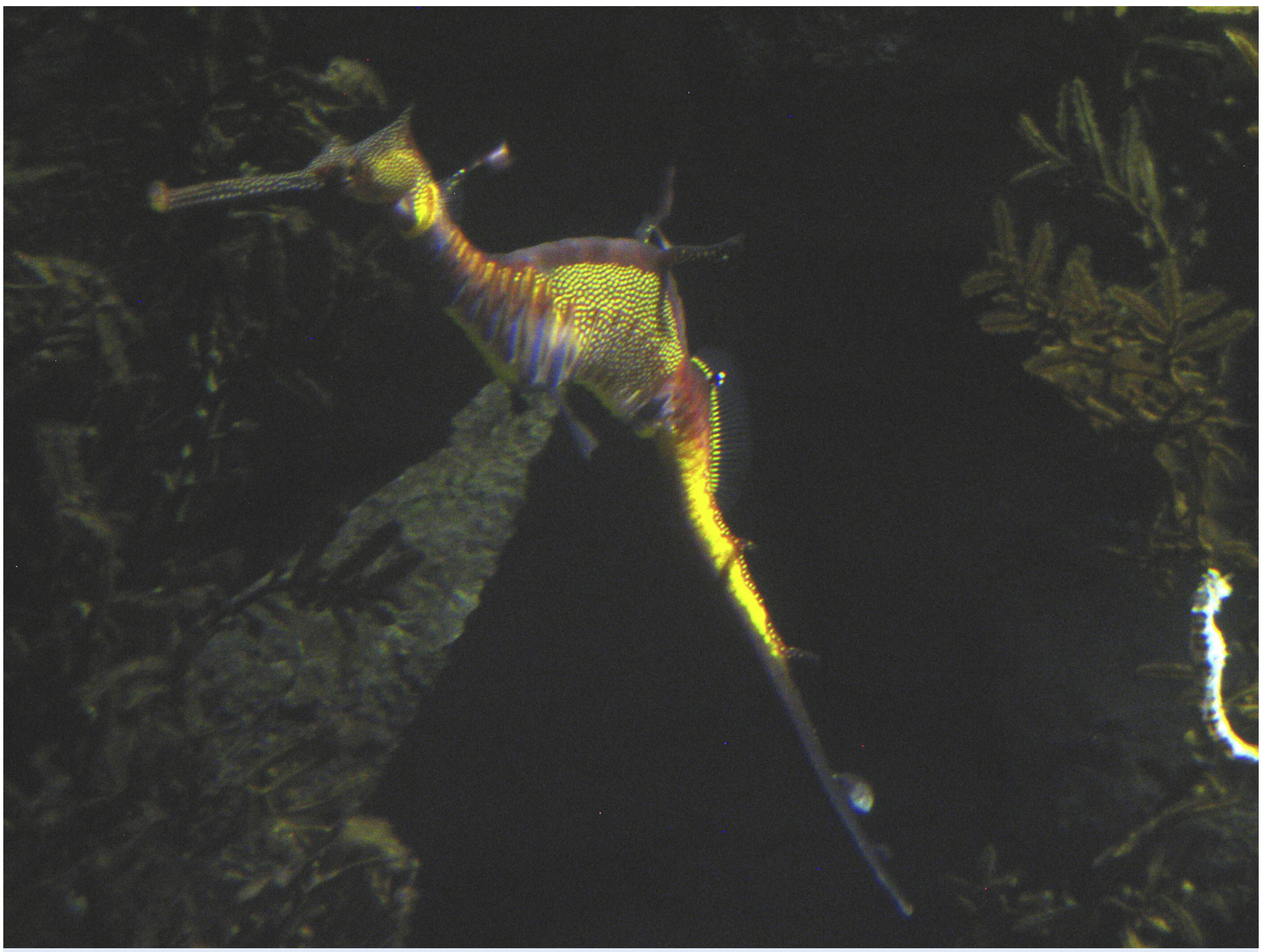

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